Efficient CV–Job Fit Classification through Distillation and Quantization

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1 Motivation and related work

HireMind is our Senior recruitment platform. We propose investigating a compact CV–job fit classifier that could later contribute evidence to its scoring workflow. The DL contribution concerns model training and efficiency; platform integration remains future work. It does not replace skill verification or recruiter decisions. A compact local classifier could reduce reliance on external API calls for matching and keep this inference step within an organization. We will measure storage and runtime; local execution alone does not guarantee privacy or lower total cost.

Gan et al. (2024) provide recruitment-screening context. BERT supplies the architecture foundation (Devlin et al., 2019); Turc et al. (2019) motivate pretrained compact students. Hinton et al. (2015) explain teacher-guided distillation, and Q8BERT (Zafrir et al., 2019) studies quantization in BERT. Our question is whether a small classifier benefits from distillation on limited CV–job pairs and retains that gain after simpler post-training quantization. These studies do not establish whether the distillation gain survives embedding-inclusive post-training quantization on our CV–job benchmark. We test that combination using a matched student architecture and held-out job descriptions, without claiming a new algorithm.

2 Question and proposed approach

Research question: Can teacher-guided training improve compact CV–job fit classification, and can quantization reduce storage while preserving the improvement?

Input is a CV and job description; output is *No Fit*, *Potential Fit* or *Good Fit*. The student is Google's pretrained BERT-Tiny checkpoint, `bert_uncased_L-2_H-128_A-2`, with a three-class head. A separately fine-tuned `bert-base-uncased` supplies teacher logits.

M0: Student trained with hard-label cross-entropy. **M1:** The same student architecture and matched initialization, trained with

$$L = 0.5L_{CE} + 0.5T^2 D_{KL}(p_T \| p_S), \quad (1)$$

where teacher and student probabilities use temperature T. **M2:** A CPU copy of M1 with dynamic INT8 Linear modules and 8-bit weight-only Embedding modules. Layer normalization and unsupported operations remain FP32. Embeddings use a dedicated weight-only configuration. M2 requires no additional training; unlike Q8BERT, we use post-training quantization.

The training strategy is Part 1; quantizing its resulting model is Part 2. The teacher is additional support, not M0. All models use identical text-pair preprocessing and a maximum of 256 tokens. Truncation frequency will be reported.

3 Hypotheses

Let F_i denote mean test macro-F1 on a 0–100 scale and S_i serialized model size.

H1: $F_1 - F_0 \geq 1.0$ percentage point.

H2: $S_2 \leq 0.40S_1$, $F_1 - F_2 \leq 1.0$ point, and $F_2 - F_0 \geq 0.5(F_1 - F_0)$, with $F_1 > F_0$.

These targets are exploratory, not pilot-derived. H2 targets at least 60% less storage: quantizing embeddings covers the dominant weight table, while metadata and remaining FP32 values limit compression. We do not promise fourfold compression or faster inference. If M1 does not improve, the improvement-retention claim fails.

4 Data and fair experiments

We use the instructor-approved resume–job-description fit dataset uploaded by cnamuangtoun on Hugging Face.[1] Its inspected revision contains 8,000 pairs: 6,241 training and 1,759 test rows,

[1] https://huggingface.co/datasets/cnamuangtoun/resume-job-description-fit

with 643 distinct CV texts and 351 job texts. Labels have counts 4,000/2,000/2,000. Six pairs have conflicting labels; quarantine their twelve rows and remove one remaining duplicate, leaving 7,987 pairs before further near-duplicate screening.

The official split shares 476 CV texts but no exact job texts. Preserve test jobs and reserve about 20% of training-job groups for validation with seed 42. Exact and whitespace-normalized checks were completed; inspect near duplicates before freezing indices. This evaluates *unseen jobs with potentially familiar CVs*, not wholly unseen candidates. The instructor approved its use for this project. Its original provenance, licence and labeling method remain undocumented. We will audit training labels, avoid redistributing CV text, and limit claims to the supplied benchmark labels.

Use AdamW, effective batch size 16 and at most three epochs. Give M0 and M1 four validation-based hyperparameter trials each, with equal update caps. Tune learning rate for M0; tune learning rate and $T \in \{2, 4\}$ for M1. Select checkpoints using validation macro-F1, then repeat selected settings at seeds 42, 43 and 44. Train the teacher on training data only; cache logits for those same examples. Freeze decisions before final test evaluation.

Report macro-F1, accuracy, per-class scores, confusion matrices, logical parameter count, serialized size, MACs/FLOPs under an explicit convention, CPU median/p95 latency and peak memory. Benchmark identical inputs, batch size, thread count and software, with 30 warm-up and 200 timed calls. Separate tokenization time and teacher-training cost. Report paired-seed variability; use job-grouped uncertainty estimates if feasible. Precision reduction need not reduce operation counts.

5 Feasibility and team plan

Completed preparation includes full-text paper review and dataset auditing. No model has been trained and no pilot performance is claimed. A Colab pilot will check checkpoint loading, embedding quantization, save/reload, finite predictions, memory, runtime and validation performance on a small subset. Free Colab is the intended starting point, subject to resource availability.

Our two-person team is approved by the instructor. Rawan leads data auditing and measurement; Hams leads training and export; both verify results and write, recording actual contributions. We selected the Technical Report track. GitHub and course-template Overleaf projects have been created and shared with the instructor. Proposed course-week allocation, subject to calendar confirmation: Week 10, labels and grouped split; Week 11, pilot, teacher and M0; Week 12, M1; Week 13, M2 and profiling; Week 14, final evaluation; Week 15, technical report, presentation and poster.

6 Risks and fallback

Undocumented data provenance and labels limit external validity; seek clarification and change data only with instructor agreement if necessary. If teacher quality is weak, report negative transfer or revise to class-weighted training before opening the test set. If embedding quantization fails compatibility checks, retain Linear-only quantization as a documented fallback and revise H2 before test evaluation; report any quality loss honestly. Reduce batch size with gradient accumulation for memory limits. Treat truncation and label ambiguity as evaluation limitations. Predictions will not support autonomous hiring or claims of demographic fairness.

7 AI use

ChatGPT and Claude assisted literature analysis, dataset research and auditing, experiment planning, workspace setup and proposal drafting. The authors are responsible for checking the content and references and for the final submission. No experimental results were generated or claimed.

References

Chengguang Gan, Qinghao Zhang, and Tatsunori Mori. 2024. Application of LLM Agents in Recruitment: A Novel Framework for Resume Screening. *arXiv:2401.08315v2*. https://arxiv.org/abs/2401.08315v2.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *NAACL-HLT*, pages 4171–4186. https://aclanthology.org/N19-1423/.

Iulia Turc, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Well-Read Students Learn Better: On the Importance of Pre-training Compact Models. *arXiv:1908.08962v2*. https://arxiv.org/abs/1908.08962v2.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. *arXiv:1503.02531*. https://arxiv.org/abs/1503.02531.

Ofir Zafrir, Guy Boudoukh, Peter Izsak, and Moshe Wasserblat. 2019. Q8BERT: Quantized 8Bit BERT. *EMC2 workshop*. DOI: 10.1109/EMC2-NIPS53020.2019.00016. https://arxiv.org/abs/1910.06188v2.